COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 6 : SLO – 1 | 1's and 2's complement
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MCQs

• 1. c — 2's complement
• 2. c — 01010 (invert all bits of 10101)
• 3. c — Flip the bits and add 1
• 4. d — Take 1's complement and mark result as negative
• 5. b — It is ignored (discarded)

Two-Mark Questions

• 1. 1's complement: invert every bit. Example: 1010 → 0101. 2's complement: invert every bit and add 1. Example: 1010 → 0101 + 1 = 0110.
• 2. Sign-magnitude: MSB is the sign, remaining bits store the magnitude; has +0 and -0; arithmetic is complex. 2's complement: negative number = 2's complement of its magnitude; single zero; addition/subtraction work with one simple adder — hence preferred.

Scenario-Based Long Answer

• 2's complement subtraction A - B = A + (2's complement of B). Example: 7 - 5 with 4 bits: A = 0111, B = 0101. 2's complement of B = 1010 + 1 = 1011. Add: 0111 + 1011 = 1 0010. A carry is generated → discard it → result 0010 = +2 (positive).
• If NO carry is generated, the result is negative and is in 2's complement form; take its 2's complement to get the magnitude and attach a minus sign. Example: 5 - 7: 0101 + 1001 = 1110 (no carry) → 2's complement of 1110 = 0010 → result = -2.

Match the Following

• 1. 1's Complement — b) Invert bits
• 2. 2's Complement — a) Add 1 to 1's complement
• 3. Sign-Magnitude — c) MSB for sign, rest for magnitude
• 4. No carry in 2's complement — d) Result is negative
• 5. Carry in 2's complement — e) Ignore carry, result is positive

Fill in the Blanks

• 1. 00110 2. 11011 (01010+1) 3. sign 4. negative (in 2's complement form) 5. end-around carry (the carry, i.e. 1, to LSB)

Scenario-Based Small Activity

• A - B = 0110 - 0011. 2's complement of B: 0011 → 1100 + 1 = 1101.
• Add: 0110 + 1101 = 1 0011. Discard the carry → 0011.
• Final result: 0011■ = 3■■ (positive, correct: 6 - 3 = 3).

True or False

• 1. FALSE — They differ (2's complement = 1's complement + 1). 2. TRUE (end-around carry).
• 3. TRUE 4. FALSE — 2's complement = invert bits THEN add 1. 5. TRUE — discard carry, result is final.